

Arda Sarp Yenicesu

Stockholm, Sweden | yenicesu@kth.se | [linkedin.com/in/arda-sarp-yenicesu](https://www.linkedin.com/in/arda-sarp-yenicesu) | asyncs.github.io

Education

KTH Royal Institute of Technology, PhD in Computer Science Jul. 2026 – Present

- Division of Robotics, Perception and Learning, School of Electrical Engineering and Computer Science
- Supervisor: Prof. Danica Kragic
- Research area: Synthetic data generation for embodied AI, Robot learning, and Long-horizon manipulation

Bilkent University, MS in Computer Engineering Jan. 2022 - Jan. 2025

- GPA: 4.0/4.0
- Relevant Coursework: Learning for Robotics, Deep Learning, Computer Vision, Machine Learning

Arizona State University, MS in Computer Science (Transfer) Jan. 2021 - Jan. 2022

- GPA: 4.0/4.0
- Relevant Coursework: Computer Vision, Natural language processing

Bilkent University, BS in Electrical and Electronics Engineering Aug. 2016 - Jun. 2020

- GPA: 3.52/4.0
- Relevant Coursework: Machine Learning, Neural Networks, Stat. Learning, Comp. Neuroscience, Algorithms

Research Experience

Doctoral Researcher, Robotics, Perception and Learning Division, KTH Royal Institute of Technology - Stockholm, Sweden Jul. 2026 – Present

- Conducting doctoral research on synthetic data generation for embodied AI, with a focus on improving the robustness and generalization of vision-language-action (VLA) models
- Investigating methods for generating reliable and diverse robot-learning data for long-horizon manipulation and deployment-oriented generalization
- Exploring planning-informed data generation and task-structure variation to improve policy learning across diverse manipulation scenarios

Graduate Research Assistant, LIRA Lab, Bilkent University - Ankara, Türkiye Jan. 2022 – Jan. 2025

- Developed novel planning heuristics for task and motion planning, inspired by humans to use auxiliary objects effectively during execution. Implemented interpretable deep learning models for robotic decision-making regarding this heuristic, enabling transparent responsibility sharing between robots and auxiliary objects
- Designed a hybrid manipulation planning framework that can achieve fine manipulation in constrained environments, that is scalable to collaborative and dynamic environments
- Contributed a locally adaptive robust anomaly detection using One-Class Classifiers to detect anomalies in sequential data sequences
- Created an experience replay methodology for fair and efficient sampling considering the whole transition history, reducing the soft divergence in continuous control tasks
- Utilized fast vision transformer architecture achieving a state-of-the-art inference speed for grasp configuration to enable real-time robotic applications in partially observable settings
- Constructed comprehensive public robotic datasets regarding anomaly detection and auxiliary object usage

Graduate Research Student, DMML, Arizona State University - Tempe, AZ Mar. 2021 – Jan. 2022

- Investigated causal inference methods for detecting suspicious user behavior in social networks
- Contributed to the development of generalized benchmarks for evaluating inference methodologies in hypergraph structures

Publications

Journal Articles

Interpretable Responsibility Sharing as a Heuristic for Task and Motion Planning <i>Yenicesu, Arda Sarp</i> [†] , Nourmohammadi, Sepehr [†] , Cicek, Berk, Oguz, Ozgur S <i>Robotics and Autonomous Systems</i>	2026
H-MaP: An Iterative and Hybrid Sequential Manipulation Planner Cicek, Berk [†] , <i>Yenicesu, Arda Sarp</i> [†] , Tuncer, Cankut Bora [†] , Demiray, Kutay, Oguz, Ozgur S. <i>IEEE Robotics and Automation Letters</i>	2025
Locally Adaptive One-Class Classifier Fusion with Dynamic lp-Norm Constraints for Robust Anomaly Detection Nourmohammadi, Sepehr, <i>Yenicesu, Arda Sarp</i> , Arashloo, Shervin Rahimzadeh, Oguz, Ozgur S <i>Pattern Recognition</i>	2025
Hungarian-Guided Policy Learning for Decentralised USV Goal Assignment and Navigation Aydin, Ahmet Gunhan, <i>Yenicesu, Arda Sarp</i> , Ergezer, Hamza <i>Under Review @ IEEE Transactions on Intelligent Vehicles</i>	2025
Beyond the Oracle: A Large-Scale Study of Subjective Reward Perception in Reinforcement Learning <i>Yenicesu, Arda Sarp</i> [†] , Aydin, Ahmet Gunhan [†] , Ceran, Elif Tugce, Ergezer, Hamza <i>Under Review @ DMLR</i>	2025

Workshop Articles

Cross-Lingual Transfer Learning for Misinformation Detection: Investigating Performance Across Multiple Languages Ozcelik, Oguzhan [†] , <i>Yenicesu, Arda Sarp</i> [†] , Yildirim, Onur [†] , Haliloglu, Dilruba Sultan [†] , Eroglu, Erdem Ege [†] , Can, Fazli <i>Workshop on Disinformation and Toxic Content Analysis @ 4th Conference on Language, Data and Knowledge</i>	2023
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Preprints

CUER: Corrected Uniform Experience Replay for Off-Policy Continuous Deep Reinforcement Learning Algorithms <i>Yenicesu, Arda Sarp</i> [†] , Mutlu, Furkan B [†] , Kozat, Suleyman S, Oguz, Ozgur S	2024
FViT-Grasp: Grasping Objects With Using Fast Vision Transformers <i>Yenicesu, Arda Sarp</i> [†] , Cicek, Berk [†] , Oguz, Ozgur S	2023

[†] Equal contribution

Teaching

Bilkent University — Artificial Intelligence • Teaching Assistant • Guest Lecturer on Deep Reinforcement Learning	Spring 2023 - Fall 2024
Bilkent University — Digital Design • Teaching Assistant	Fall 2022 - Spring 2023
Bilkent University — Computer Networks • Teaching Assistant	2022–2024 (Spring terms)

Professional Experience

AI Engineer , ASELSAN - Ankara, Türkiye	Apr. 2025 – Feb. 2026
- Developed autonomous task and motion planning algorithms for unmanned surface vehicles, adapting academic research into defense-grade, real-world platforms. - Designed sequential decision-making frameworks for multi-agent robotic systems, collaborating with multidisciplinary teams on mission-critical AI applications.	

Skills

Languages: Turkish (native), English (fluent)

Programming: Python, C/C++, Java, MATLAB, SQL

Domains: Robotics, Task and Motion Planning, Reinforcement Learning, Deep Learning, Machine Learning, Computer Vision

Technologies: PyTorch, TensorFlow, Keras, RAI, Stable Baselines, Gym, OpenAI API, Pinecone, Git, Docker